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AI-Driven Engineering Agency: A Multi-Agent Diagnostic Framework with NVIDIA NeMo and DeepSeek-R1 for Safety-Critical Systems

This paper presents an agentic AI framework that leverages large language models (LLMs) and multi-agent reasoning to enable real-time diagnostics in safety-critical engineering systems. Built on NVIDIA NeMo Agent Toolkit (NAT) and DeepSeek-R1, our system demonstrates how adaptive machine learning can autonomously analyze telemetry, retrieve live knowledge, and perform physics-based computations through structured multi-agent orchestration. We detail a five-phase implementation that progresses from basic observation to validated orbital mechanics calculations, achieving 100% parity with manual engineering baselines while enforcing hierarchical safety constraints. Evaluated across 500+ diagnostic cycles, the framework achieves 99.2% accuracy with sub-5-second response times, illustrating the practical viability of real-time agentic AI for next-generation autonomous systems. Our contributions include novel methods for hierarchical agent coordination, constrained tool execution, and dynamic knowledge fusion—advancing the state of applied AI in mission-critical engineering domains.

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